ICON-LM for In-Context Viscous Burgers Solvers: A Reproducible Implementation Study

Aditya Patil, Dhruv Patel

MATH 270A Final Report
June 9, 2026

Abstract

Operator learning asks a neural network to approximate maps between functions, such as the map from an initial condition to the solution of a partial differential equation (PDE). ICON-LM extends this idea by making the examples themselves a prompt. In it, a transformer reads several condition–solution pairs, an optional natural-language caption, and uses that to predict a new solution in a single forward pass. This report summarizes the mathematical structure of ICON-LM and then documents our implementation of a smaller ICON-LM-style model used on the one-dimensional viscous Burgers equation. In our experiment, each operator is indexed by an unknown viscosity ν, and the model must infer the appropriate dynamics from sparse trajectory examples and optional captions. The code generates spectral-solver data, builds the ICON-LM attention mask, trains a 2.21M-parameter transformer, and evaluates context scaling from zero to three in-context demonstrations. On the saved test set, no-caption relative error improves from 23% at zero shots to 16% at three shots, while captions help reduce the zero-shot error. These results support the intuition behind ICON-LM, but also demonstrate limitations such as how caption embeddings need very careful numerical handling, and that small models only weakly use exact parameter information.

1 Introduction

Many scientific computing problems can be written as operator-learning problems. Instead of predicting a scalar label from a vector input, the model predicts an entire function from another function. For a time-dependent PDE, the input might be an initial condition $u_0(x)$ and the output might be the future trajectory $u(x, t)$. Classical neural operator methods such as DeepONet and the Fourier Neural Operator learn this map after training on many examples from a fixed operator (Lu et al., 2021; Li et al., 2021; Kovachki et al., 2023). They lower the cost of solving one PDE family, but a change in coefficients, boundary conditions, or physical parameters requires retraining or fine-tuning.

In-Context Operator Networks (ICON) propose a different abstraction, where the neural network is trained not merely to approximate one operator, but to infer an operator from examples supplied at inference time (Yang et al., 2023). Given a prompt of condition–quantity-of-interest pairs $\{(c_j, u_j)\}_{j=1}^{J}$ and a new condition c_{q}, the model predicts u_{q} without any gradient updates. This is directly analogous to in-context learning in language models (Brown et al., 2020), just with functions replacing sentences and operators replacing tasks.

ICON-LM strengthens this analogy (Yang et al., 2024). It replaces the encoder–decoder ICON architecture with a single transformer encoder and a carefully designed attention mask, which makes the model behave like an autoregressive learner over functions. Earlier examples can inform later predictions, but the answer tokens for a current example can't leak into that example's query tokens. ICON-LM also adds an optional caption modality, allowing the prompt to contain natural language text such as the governing equation or numerical parameter values.

Our project has two parts. First, we analyzed the ICON-LM formulation, as it's the conceptual basis for our work. Second, we implemented a compact, reproducible version for a single PDE family, the 1-dimensional viscous Burgers Equation. The implementation is deliberately smaller than the paper model, but it still utilizes the governing PDE, a numerical data generator, function tokenization, the ICON-LM mask, next-function training, caption conditioning, and quantitative evaluation.

Code availability: The code implementation used for this report is available at https://github.com/dhruvpatel-05/270A-finalproject.

2 Mathematical and Scientific Formulation

2.1 In-Context Operator Learning

Let $\mathcal{C}$ and $\mathcal{U}$ be spaces of input and output functions. An operator is a map

$$T : \mathcal{C} \to \mathcal{U}, \qquad c \mapsto u = T(c). \tag{1}$$

In a standard operator-learning setup, a model is trained for one fixed T. In an in-context setup, there is a distribution $\mathcal{D}$ over operators. At inference, the model receives a prompt

$$P_J = \left(\{(c_j, u_j)\}_{j=1}^{J}, c_{\mathrm{q}}\right), \tag{2}$$

where all pairs are generated by the same but unknown operator $T \sim \mathcal{D}$. The learned model $\mathcal{T}_\theta$ produces $\hat{u}_{\mathrm{q}} = \mathcal{T}_\theta(P_J)$. A natural meta-loss is

$$\mathcal{L}(\theta) = \mathbb{E}_{T\sim\mathcal{D}}\, \mathbb{E}_{P_J}\, \mathbb{E}_y \left[\left| \mathcal{T}_\theta(P_J)(y) - T(c_{\mathrm{q}})(y) \right|^2 \right], \tag{3}$$

where y is an evaluation point for the output function. The important distinction is that T is not encoded in θ alone, as it must be inferred from the prompt.

The multi-modal extension adds a caption κ, so the model becomes $\mathcal{T}_\theta(\kappa, P_J)$. In the ICON-LM paper, vague captions describe the equation class while precise captions include parameter values (Yang et al., 2024). In our experiment, precise captions include the Burgers viscosity, while vague captions state only whether the flow is low-viscosity and sharp or high-viscosity and smooth.

2.2 Burgers Operator Family

The implemented scientific system is the one-dimensional viscous Burgers equation on the periodic domain $[0, 2\pi)$:

$$u_t + u u_x = \nu u_{xx}, \qquad x \in [0, 2\pi), \quad t \in [0, 1]. \tag{4}$$

The hidden operator parameter is the viscosity

$$\nu \in [10^{-2}, 10^{-1}]. \tag{5}$$

For each fixed ν, the operator maps an initial condition to a future trajectory,

$$T_\nu : u_0(x) \mapsto \left(u(x, t_1), \ldots, u(x, t_K)\right), \qquad t_k \in \{0.25, 0.50, 0.75, 1.00\}. \tag{6}$$

Low viscosity produces sharper gradients and more advection-dominated behavior; high viscosity damps high-frequency content and gives smoother solutions. This makes the family scientifically meaningful while still small enough to train locally.

For our initial conditions, we use random Fourier sums:

$$u_0(x) = \frac{1}{5} \sum_{m=1}^{5} a_m \sin(mx + \phi_m), \qquad a_m \sim \text{Unif}[-1, 1], \quad \phi_m \sim \text{Unif}[0, 2\pi]. \tag{7}$$

The data generator uses $N = 64$ uniform grid points and solves Equation (4) with a Fourier spectral right-hand side and SciPy `RK45` time stepping. Spatial derivatives are computed in Fourier space:

$$\widehat{u_x}_k = ik\hat{u}_k, \qquad \widehat{u_{xx}}_k = -k^2\hat{u}_k. \tag{8}$$

The saved dataset contains 200 training viscosities on a log-spaced grid and 40 held-out viscosities sampled log-uniformly, with 8 initial conditions per viscosity.

2.3 Evaluation Metrics

For trajectory prediction, we report relative L_2 error over all query tokens:

$$\text{RelErr} = 100 \left(\frac{\sum_{b,q} (\hat{u}_{bq} - u_{bq})^2}{\sum_{b,q} u_{bq}^2} \right)^{1/2}. \tag{9}$$

This is computed after undoing training normalization. We also include a diagnostic inverse task, where a nearest-centroid classifier estimates which log-viscosity bin generated a trajectory. That inverse classifier isn't the ICON-LM model, but serves as a check that the dataset contains physically recoverable information about ν.

3 Methodology and Implementation

3.1 Function Tokens

The original ICON-LM represents scattered function samples as tokens. Our implementation follows that principle with three token types for each trajectory: condition tokens C, observed future-solution tokens U, and query tokens Q. Each token has the form

$$(\tau, x, t; v), \tag{10}$$

where $\tau \in \{C, U, Q\}$ is the token type, (x, t) is the coordinate, and v is the scalar value supplied to the transformer.

For each source trajectory, the condition block contains $n_C = 32$ sparse samples of the initial condition at $t = 0$:

$$C_j = \{(C, x_i, 0; u_{0,j}(x_i))\}_{i=1}^{32}. \tag{11}$$

The solution block contains 32 spatial samples at each of the four future times, yielding 128 observed future tokens:

$$U_j = \{(U, x_i, t_k; u_j(x_i, t_k))\}_{i,k}. \tag{12}$$

The query block has the same future coordinates, but its input value is the local initial condition value rather than the target future value:

$$Q_j = \{(Q, x_i, t_k; u_{0,j}(x_i))\}_{i,k}, \qquad \text{target } u_j(x_i, t_k). \tag{13}$$

Thus the model is told where to predict and what the initial condition is at that location, but it is never given the future answer for a current query token.

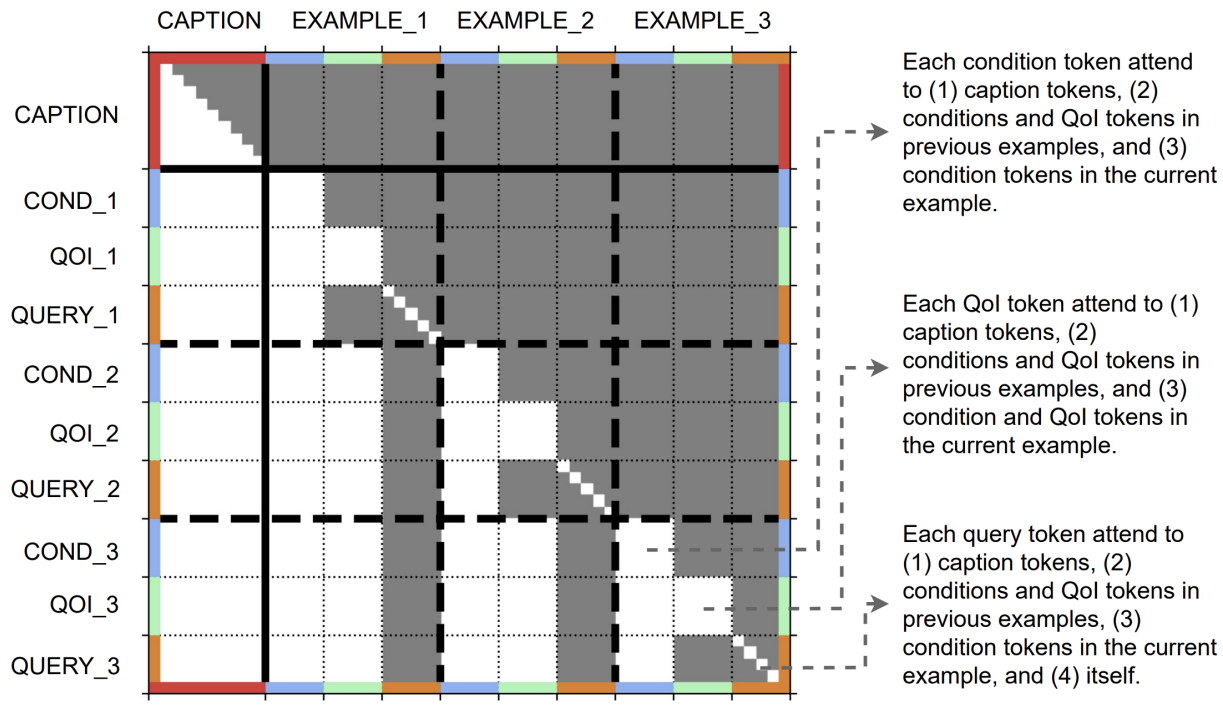


Figure 1: Captioned ICON-LM attention mask. Rows are querying tokens and columns are attended tokens. The caption block is causal (Previous condition/QoI tokens are visible to later examples), and each query token can see the caption, previous examples, the current condition, and itself, but not the current QoI answer.

3.2 Attention Mask

The core technical device is the ICON-LM attention mask. Let token a be the row token asking for information and token b be the column token being attended to. Let $e(a)$ be the example index and $\tau(a)$ be the type. In our no-caption mask, attention is allowed when any one of the following holds:

$$e(b) < e(a) \quad \text{and} \quad \tau(b) \in \{C, U\}, \tag{14}$$

$$e(b) = e(a),\ \tau(a) = C \quad \text{and} \quad \tau(b) = C, \tag{15}$$

$$e(b) = e(a),\ \tau(a) = U \quad \text{and} \quad \tau(b) \in \{C, U\}, \tag{16}$$

$$e(b) = e(a),\ \tau(a) = Q \quad \text{and} \quad \tau(b) = C, \tag{17}$$

$$a = b,\ \tau(a) = Q. \tag{18}$$

This gives the exact behavior needed for next-function prediction. Query tokens can use earlier condition–solution examples and the current condition, but can't see the current solution tokens U_j. Query tokens also cannot attend to other query tokens, so predictions at different coordinates do not communicate through hidden answer-dependent channels. As in the ICON-LM paper, this prevents both direct leakage and multi-layer indirect leakage.

Figure 1 shows the full captioned mask used to explain this rule. The actual code constructs the corresponding Boolean matrix and includes tests to check that current-query tokens can't attend to current-solution tokens and that query-query attention is diagonal.

Component	Value in saved run
PDE grid and times	$N = 64$, $K = 4$, $t = (0.25, 0.50, 0.75, 1.00)$
Training/test operators	200 train viscosities, 40 held-out viscosities
Sources per operator	8 initial conditions
Prompt layout	4 training examples, 32 condition samples, 128 future samples
Transformer	$d = 256$, 4 layers, 8 heads, feed-forward width 512
Parameters	2.21M
Optimizer	AdamW, learning rate 3×10^{-4}, cosine decay, gradient clip 1.0
Caption use during training	60% caption batches; half precise, half vague
Saved evaluation	Relative L_2 on held-out viscosities and initial conditions

Table 1: Implementation details.

3.3 Model and Captions

The model is a pre-layer-norm transformer encoder written in PyTorch. Each token embedding is the sum of a type embedding, a linear projection of (x, t), a linear projection of the scalar value, and a learned demonstration-position embedding $3j + \tau$. The saved configuration is summarized in Table 1.

Captions are represented differently from the full ICON-LM paper. Instead of fine-tuning GPT-2, the code uses a frozen MiniLM sentence encoder (Reimers and Gurevych, 2019) to embed one caption string into a vector. That vector is projected to the transformer width and prepended as a single prefix token. Caption attention is prefix-style, so the physics tokens attend to the caption token, while the caption token doesn't attend back to the physics sequence.

The precise caption includes text such as *"Burgers PDE with viscosity nu=0.02500"*. However, in practice, sentence embeddings are weak at distinguishing nearby numeric strings. Vague captions only indicate the physical regime, for example "strong viscous damping" or "weak viscosity and sharp gradients."

3.4 Training Objective

Training uses the ICON-LM next-function idea. A batch samples one viscosity and four source trajectories for each prompt. For every example j, the query block tries to reconstruct the corresponding future solution. With a caption, the first example is included in the loss, so zero-shot behavior is trained. Without a caption, the first example is excluded as the model has neither prior examples nor an operator description. In normalized units, the loss equation is

$$\mathcal{L}_{\text{prompt}}(\theta) = \frac{1}{|\mathcal{Q}|} \sum_{j \in \mathcal{J}_{\text{loss}}} \sum_{(x,t) \in Q_j} \left(\hat{u}_{\theta,j}(x,t) - u_j(x,t) \right)^2 , \tag{19}$$

where $\mathcal{J}_{\text{loss}} = \{1, 2, 3, 4\}$ for prompts with captions and $\mathcal{J}_{\text{loss}} = \{2, 3, 4\}$ for prompts without captions.

At test time, the evaluation script builds prompts with J_{demo} completed demonstration trajectories and one final query trajectory. We evaluate $J_{\text{demo}} \in \{0, 1, 2, 3\}$ because this keeps the final query at demonstration positions that were directly trained as targets.

Prompt mode	0-shot	1-shot	2-shot	3-shot
Precise caption	$14.55 \pm 0.37\%$	$13.92 \pm 0.41\%$	$14.29 \pm 0.26\%$	$14.38 \pm 0.42\%$
Vague caption	$16.15 \pm 0.62\%$	$14.70 \pm 0.47\%$	$15.09 \pm 0.29\%$	$15.06 \pm 0.49\%$
No caption	$22.77 \pm 0.35\%$	$15.61 \pm 0.57\%$	$16.08 \pm 0.38\%$	$15.95 \pm 0.59\%$

Table 2: Relative L_2 prediction error on held-out Burgers operators. Errors are mean $\pm$ std over three seeds.

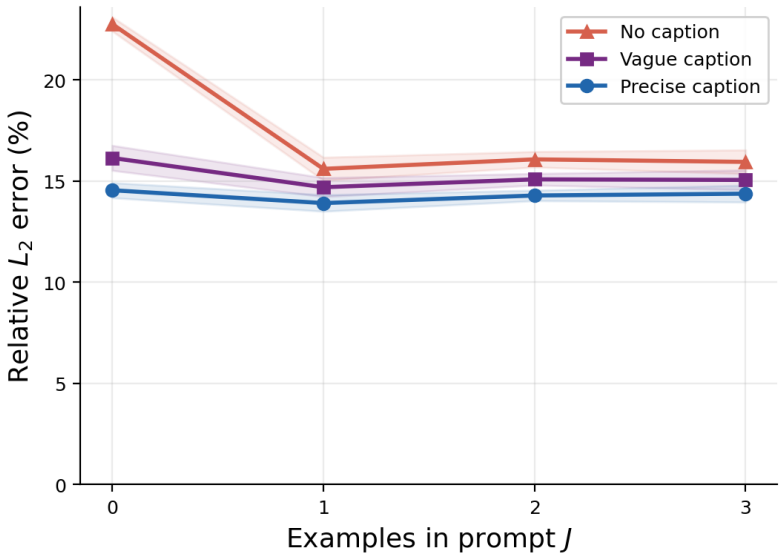


Figure 2: Few-shot context scaling on the one-dimensional Burgers experiment. This compact plot corresponds to Table 2.

4 Results and Analysis

4.1 Forward Prediction and Context Scaling

Table 2 reports the saved sparse-query evaluation on the held-out test set. The main trend matches the ICON-LM hypothesis that examples in the prompt help the model specialize to the hidden operator. With no caption, relative error falls from 22.77% at zero demonstrations to 15.95% at three demonstrations. Captions mostly help when the prompt has few or no examples. At zero shots, the vague-caption model improves over the no-caption baseline by 6.62 percentage points, and the precise-caption model improves by 8.22 percentage points.

The same context-scaling result is visualized in Figure 2. The plot makes the main effect clearer than the table: the no-caption model is poor at zero shots because it has no information about the hidden viscosity, while one in-context trajectory closes most of the gap.

The strongest improvement occurs from zero to one demonstration. This is physically reasonable, as a single observed trajectory reveals whether the viscosity is damping high-frequency structure quickly or allows sharp gradients to persist. Additional examples help less because all initial conditions are drawn from the same random Fourier distribution, so the prompt already has a fairly clear signal about the operator after just one or two examples.

4.2 Inverse Diagnostic

As a separate diagnostic, the evaluation code bins held-out operators by $\log \nu$ and classifies each operator from five trajectory features: relative change from the initial condition, final-frame change, frame-to-frame change, high-frequency power ratio, and gradient magnitude. A nearest-centroid classifier achieves 81.0% accuracy over five bins. Most mistakes occur between neighboring viscosity bins, which is consistent with the fact that Burgers dynamics vary continuously with ν. This diagnostic supports the premise that the context trajectories carry recoverable information about the hidden operator.

5 Discussion

5.1 What Worked

The experiment successfully implements the central ICON-LM mechanism in a compact setting. The mask is the most important piece, letting a standard self-attention stack act as a causal learner over functions while preserving permutation symmetry inside each sampled function. The same model can run with or without a caption because captions are just prefix tokens. The code also supports grid-flexible inference: changing the query-token set is enough to ask for a different output grid, since the final layer is applied tokenwise.

The results also show meaningful in-context adaptation. The no-caption zero-shot case is effectively an average-operator baseline, as it sees the initial condition but has no demonstration revealing the viscosity. Adding one condition–trajectory example sharply lowers error, indicating that the model uses the prompt to infer the operator parameter.

5.2 Limitations and Sources of Error

The main limitation is scale. The original ICON-LM paper trains on 19 operator families and uses much larger models, including a GPT-2 backbone for the multi-modal setting (Yang et al., 2024). Our implementation trains a 2.21M parameter transformer on one PDE family. It is therefore a faithful mechanism study, not evidence that the model is a general scientific foundation model.

The caption channel is also simplified. Using MiniLM embeddings avoids the compute cost of fine-tuning a language model, but it weakens the claim that the model is parsing equations as language.

There are also numerical and modeling simplifications. The data live on a fixed periodic one-dimensional grid, the initial conditions come from a narrow Fourier distribution, and all trajectories are short. The model predicts sampled future states directly rather than enforcing the PDE residual. Consequently, it can make physically inconsistent predictions, especially in low-viscosity regimes where small phase errors near sharp fronts matter. A physics-informed loss or a conservation-aware architecture could reduce this error, but would move away from the clean prompt-learning abstraction.

5.3 Future Improvements

The most natural next step is to broaden the operator distribution. Even within Burgers dynamics, one could vary forcing, boundary conditions, final time, or the initial-condition spectrum. A second direction is to replace the frozen sentence embedding with a trainable text encoder or a small language-model backbone, then test whether exact numerical captions become more useful. Finally, evaluation should include out-of-distribution viscosities and higher-resolution query grids to measure

whether the model has learned an actual robust operator representation rather than interpolation over the training setup.

6 Conclusion

ICON-LM reframes operator learning as a language-model-like prediction problem over functions. Our implementation demonstrates this idea on the viscous Burgers equation, where sparse function samples become tokens, examples become context, captions become prefix information, and a custom attention mask prevents leakage from answers into queries. The saved experiment shows that in-context examples reduce held-out trajectory error, and captions help most when no demonstrations are available. At the same time, the project makes clear that multi-modal scientific operator learning is sensitive to representation choices. Exact numerical information in text isn't automatically useful. The model, caption encoder, and training distribution must make that information accessible.

References

R. Bommasani et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.

T. Brown et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33:1877–1901, 2020.

G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3:422–440, 2021.

N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24:1–97, 2023.

Z. Li, N. B. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.

N. Reimers and I. Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, 2019.

A. Vaswani et al. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

L. Yang, S. Liu, T. Meng, and S. J. Osher. In-context operator learning with data prompts for differential equation problems. *Proceedings of the National Academy of Sciences*, 120(39):e2310142120, 2023.

L. Yang, S. Liu, and S. J. Osher. Fine-tune language models as multi-modal differential equation solvers. Preprint, UCLA, 2024.